

organism enhanced by everything technology has to offer.

In the U.S. Future Force combat systems are being tested and designed to make the 21st century American cyborg soldier a more effective instrument of war; a veritable cyborg able to communicate with augmented cognition more speedily and efficiently. The U.S. military is funding projects to integrate human with artificial intelligence.



Cyborg Soldiers

- ▶ **Problem:** Human brains are superior to computers at visual recognition but inferior at information processing.
- ▶ **Solution:** Human-machine integration.
- ▶ **Human component:** A soldier or analyst who scans scenes or images.
- ▶ **Machine component:** Sensors that monitor the brain's activity and relay information about it to commanders or computers.



Combining Man and Machine

The 'Brain Interface Program' is one of the most lavishly funded of nearly all the DARPA bioengineering efforts (the project has been given over \$24 million budget). It is aimed at developing ways to 'integrate' soldiers into machines -literally- by wiring them (remotely or directly) to their planes, tanks, or computers. An implantable brain chip has been implemented as well via the integration of stimulus-response signals in the brain via electrodes. In fact it is often said that the human is becoming the weakest link in defense systems.

Combining man and machine to enhance innate soldier capabilities is the hallmark of a soldier-cyborg transformation. Increasing the man-machine interface in the unpredictable environment of war has enormous potential to change the human dimension of war. The immediate future promises to hold some very interesting prospects for cyborg soldiers.

Cybernetics

Human Cybernetics in the field of life extension, refers to how technology has, is and will evolve to support the ideology of extending human lifespan. Scientists already possess prosperous knowledge in how to avoid some degenerative dis-